Possible Problems for Quiz 2 to be take 1/24/19. Of these problems I will choose 3 to be on the quiz.

Problem 1

Transform the following matrix to echelon form

$$\begin{bmatrix} 1 & -2 & 3 \\ 2 & -3 & 5 \\ -1 & 4 & -5 \end{bmatrix}$$

Problem 2

Suppose we want to find the 2nd degree polynomial that goes through the points (1,4) , (2,7), (3,14)

Set up the linear system you would use to find the coefficients for $f(x) = ax^2 + bx + c$. Don't solve it.

Problem 3

For the augmented matrix below, circle the pivot positions and state how many free parameters the system has.

$$\left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & 6 & 8 & 2 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

Problem 4

Balance the following chemical equation

$$x_1 C_8 H_{18} + x_2 O_2 \rightarrow x_3 CO_2 + x_4 H_2 O$$

Problem 5

Let $\vec{u} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ find $\vec{u} + \vec{v}$ and draw all three vectors.

Problem 6

Find a linear combination of $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \vec{v} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ such that $c_1\vec{u} + c_2\vec{v} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$, that is, what are c_1, c_2?